BLOCKCHAIN-BASED CROSS-BORDER PAYMENTS

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

Use Case Report

submitted by

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CERTIFICATE

This is to certify that the Use Case report entitled **"Blockchain-Based Cross-Border Payments"** that is being submitted by **Anisetty Bhavitha (23501A0509)**, as part of Assignment-2 for the **Blockchain Technology (23CS4701B)** course during the academic year **2026-27**.

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1. INTRODUCTION

1.1. Overview of Blockchain Technology

Blockchain is a shared digital record book. Many computers (called nodes) keep the same copy of the record, so no single person or company controls it. Every new set of records is stored in a block, and each block carries the hash (a unique digital fingerprint) of the block before it. Because of this link, changing old data is almost impossible without everyone noticing. Blockchain started with digital money [1], but today it is used in many areas such as banking, supply chains, healthcare and payments.

1.2. Relevance of Blockchain in Cross-Border Payments

Sending money from one country to another is slow and costly today. A payment often passes through several banks before it reaches the receiver. Each bank keeps its own records, charges its own fee and takes its own time. The sender usually cannot see where the money is at a given moment.

A blockchain-based cross-border payment system keeps one shared record of every payment. All approved banks and payment providers can see the same data, so checking and matching records becomes much easier. Smart contracts can apply the payment rules automatically, and the status of each payment can be traced from start to end. This makes international transfers more transparent, traceable and efficient.

In this report, a permissioned (controlled) blockchain is used, where only approved banks and payment providers can take part. This suits money transfers because the data is sensitive and the rules of different countries must be followed. The money itself is still settled through the normal banking channels, while the blockchain keeps the trusted record of the payment.

2. BACKGROUND

Traditional cross-border payments depend on a chain of banks and on manual checking. This creates delays, extra cost and confusion. Below are the main problems in the present system:

2.1. Many Intermediaries

A payment often moves through correspondent banks before it reaches the final bank. Every bank in the chain adds time and charges a fee, and each one must trust the others' records.

2.2. Slow Settlement

Payments can take one or more working days to reach the receiver. Bank holidays, different time zones and manual approvals make the delay longer.

2.3. High Fees and Currency Conversion Costs

Senders pay transfer fees, intermediary fees and exchange-rate margins. It is often hard to know the final amount the receiver will get.

2.4. Lack of Transparency and Tracking

Once a payment is sent, the sender and the receiver usually cannot see its exact stage. They must contact the bank to ask about it, which wastes time and reduces trust.

2.5. Difficult Reconciliation

Every bank keeps its own ledger. At the end of the day, banks must match their records with each other. Differences in these records cause errors and need manual correction.

2.6. Fraud and Errors

Wrong account details, duplicate payments and fraud attempts are real risks. Since the information is spread across many systems, such problems are hard to find and fix quickly.

2.7. Compliance Burden

Each country has its own rules for identity checks, anti-fraud checks and reporting. Banks must follow these rules for every transfer, which increases the work and the processing time.

2.8. Limited Access

Small businesses and people in remote areas may have limited access to fast and low-cost international payment services.

Because of these problems, there is a need for a payment system that is faster, clearer and easier to audit. Blockchain can help by giving all participants a shared, tamper-resistant record of each payment.

3. BLOCKCHAIN BASICS

Blockchain is a distributed ledger that is secure, transparent and hard to change. The key ideas behind it are explained below.

3.1. Decentralization

In a normal system, one central company or bank controls the database. If it fails or is hacked, the whole system suffers. In a blockchain, many nodes keep the same data and check each other. No single party has full control, and the system keeps working even if one node fails.

3.2. Immutability

Once a transaction is written to a block and accepted by the network, it cannot be quietly changed or deleted. Each block stores the hash of the previous block, so changing one block would break all the blocks after it. This gives a trusted history, which is very useful for payment records and audits.

3.3. Transparency

All approved participants can see the same copy of the records. This removes hidden changes and helps everyone agree on the same facts. In a permissioned network, the data is visible only to the approved members.

3.4. Types of Blockchain

Blockchains are mainly of four types, as shown in Table 3.1.

Blockchain Type	Meaning	Typical Use
Public Blockchain	Anyone can take part according to the network rules.	Open blockchain networks
Private Blockchain	Controlled by one organization.	Enterprise payment systems
Consortium Blockchain	Controlled by several trusted organizations.	Banks and payment providers
Hybrid Blockchain	Combines public and private features.	Controlled and transparent systems

Table 3.1: Types of Blockchain

Why permissioned / consortium? Banks and payment institutions can decide which organizations, users and validator nodes are trusted and allowed to join. This gives them control over privacy and rules, which is needed for cross-border payments.

3.5. Smart Contracts

A smart contract is a program stored on the blockchain. It runs automatically when its function is called and its conditions are met [2]. In this project, smart contracts check who is

allowed to send a payment, record the payment and update its status, so no manual step is needed for these tasks.

3.6. Consensus Mechanisms

A consensus mechanism is the method used by the network to agree on which transactions are valid. The common ones are:

- **Proof of Work (PoW)** – Nodes solve difficult puzzles to add a block. It is secure but uses a lot of energy [1].
- **Proof of Stake (PoS)** – Validators are chosen based on the value they have staked.
- **Proof of Authority (PoA)** – Only approved validators confirm transactions. It is fast and suitable for permissioned networks [3].

3.7. Cryptographic Security

Blockchain uses cryptography to protect data. Each user has a public key and a private key. A transaction is signed with the private key, so everyone can check that it came from the right user. Hash functions turn data into fixed-size codes that cannot be reversed, which keeps the chain of blocks safe.

4. USE CASE OVERVIEW

The Blockchain-Based Cross-Border Payments system is designed to make international money transfers more transparent, traceable and efficient. Blockchain records the payment transactions and helps coordinate settlement, which reduces the effort of matching records between banks.

4.1. Objectives

- **Transparency:** Authorized parties can see the status of a payment at any time.
- **Traceability:** Every payment has a record that can be followed from start to end.
- **Efficiency:** Smart contracts apply the payment rules automatically and reduce manual checking.
- **Less reconciliation work:** Banks use one shared record instead of matching many separate ledgers.
- **Security:** Only registered users and approved institutions can perform actions in the system.

4.2. Scope of the System

The system includes the following actors:

1. **Sender / Payer:** Starts a cross-border payment, enters the beneficiary and amount details, and views the payment status.
2. **Bank / Payment Service Provider:** Performs payment checks and supports settlement or currency conversion where needed.
3. **Beneficiary / Receiver:** Receives the payment and can view the transaction status.
4. **Blockchain Network:** Validates transactions and keeps the shared payment record.

4.3. System Architecture

The architecture of the system is shown in Figure 4.1. It has three main layers, and traditional payment rails are connected for currency conversion and final settlement.

4.3.1. User Layer

- Senders log in, enter payment details and track the status.
- Banks and payment providers check payments and confirm settlement.
- Beneficiaries view the payment status and receive the funds.

4.3.2. Web Application Layer

- **Frontend (web or mobile app):** The screen through which users interact with the system.
- **Backend / API layer:** Handles login, checks the input and talks to the blockchain.

4.3.3. Blockchain Layer

- **Smart contract:** Holds the payment rules and stores the payment details and status.
- **Consensus (Proof of Authority):** Authorized validators confirm each transaction.
- **Blockchain network:** Stores the payment records so that they are hard to change.

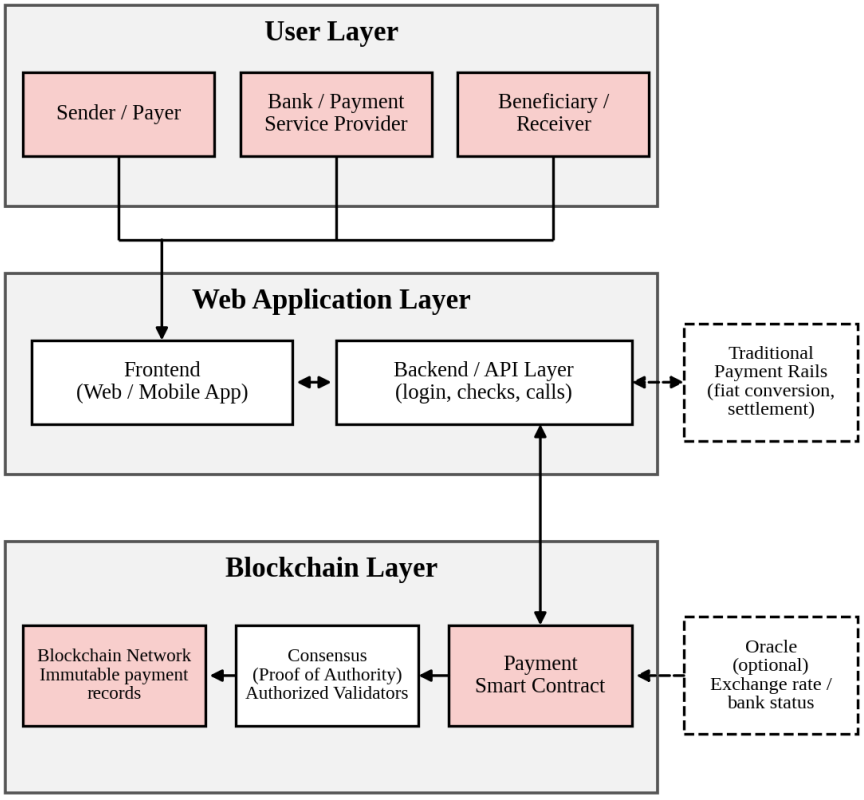


Figure 4.1: Architecture of Blockchain-Based Cross-Border Payments System

4.4. Cross-Border Payment Transaction

When a payment is submitted, the transaction can contain:

- Sender ID / wallet or account reference
- Beneficiary ID / wallet or account reference
- Source and destination countries or currencies
- Payment amount
- Transaction date and time
- Transaction status
- Payment / institution reference

In simple words, a payment transaction is the blockchain record created when the sender starts an authorized cross-border transfer.

5. IMPLEMENTATION

5.1. Setting Up the Blockchain Environment

The smart contract is written in Solidity (version 0.8.19) for an Ethereum-compatible blockchain. During development, Truffle and Ganache are used to build and test the contract on a local test blockchain, and MetaMask is used to manage test accounts. For the final permissioned network, an Ethereum-compatible client such as Hyperledger Besu can be used with Proof of Authority, so that only approved bank nodes act as validators [3][5].

```
// 2_deploy_contracts.js (Truffle migration)
const CrossBorderPayments = artifacts.require("CrossBorderPayments");

module.exports = function (deployer) {
      deployer.deploy(CrossBorderPayments);
};
```

5.2. Defining the Smart Contract

5.2.1. Data Structure

The contract keeps a Payment structure that stores the payment ID, sender, beneficiary, destination country, currency, amount, creation time and current status. The status can be Initiated, Validated, Settled or Rejected. Three mappings store the registered participants, the approved banks and the payments.

```
enum Status { Initiated, Validated, Settled, Rejected }

struct Payment {
    uint256 id;
    address sender;
    address beneficiary;
    string  currency;
    uint256 amount;
    Status  status;
}
```

```
address public admin;
uint256 public nextPaymentId = 1;
mapping(address => bool) public participants;
mapping(address => bool) public banks;
mapping(uint256 => Payment) public payments;

event PaymentInitiated(uint256 id, address sender, address beneficiary, uint256 amount);
event StatusUpdated(uint256 id, Status status);
event PaymentSettled(uint256 id);

modifier onlyAdmin() { require(msg.sender == admin, "Only admin"); _; }
modifier onlyParticipant() { require(participants[msg.sender], "Not registered"); _; }
modifier onlyBank() { require(banks[msg.sender], "Only approved bank"); _; }

constructor() { admin = msg.sender; }
```

5.2.2. Managing Participants

Only the administrator can add participants. A participant can be marked as a bank or payment provider. The verifyParticipant function lets anyone check whether an address is registered. This keeps unknown users out of the payment network.

```
function addParticipant(address user, bool isBank) external onlyAdmin {
    participants[user] = true;
    if (isBank) { banks[user] = true; }
}

function verifyParticipant(address user) external view returns (bool) {
    return participants[user];
}
```

5.2.3. Initiating a Payment

A registered sender calls the initiatePayment function with the beneficiary address, destination country, currency and amount. The contract checks the rules (for example, the beneficiary must be registered and the amount must be more than zero). If everything is correct, it creates a payment with a unique ID and the status “Initiated”.

```
function initiatePayment(address beneficiary,
   string memory currency, uint256 amount)
   external onlyParticipant returns (uint256) {
   require(participants[beneficiary], "Beneficiary not registered");
   require(amount > 0, "Invalid amount");
   // store the payment with status Initiated
   return nextPaymentId++;
}
```

5.2.4. Viewing Payment Status and Details

Users can read the current status or the full details of a payment by using its ID. These are read-only functions, so they do not change the blockchain and cost no gas when called from outside.

```
function getPaymentStatus(uint256 id) external view returns (Status) {
    return payments[id].status;
}

function getPayment(uint256 id) external view
    returns (address, address, string memory, uint256, Status) {
    Payment memory p = payments[id];
    return (p.sender, p.beneficiary, p.currency, p.amount, p.status);
}
```

5.2.5. Updating Status and Confirming Settlement

Only an approved bank can update a payment. After the bank checks the payment, it marks it as Validated or Rejected using updateStatus. Once the money is settled through the normal banking channel, the bank calls confirmSettlement and the status becomes Settled. The contract stores the record of the payment; the actual money moves through the supported payment rails.

```
function updateStatus(uint256 id, Status newStatus) external onlyBank {
    require(payments[id].status == Status.Initiated, "Already processed");
    require(newStatus == Status.Validated || newStatus == Status.Rejected,
        "Invalid status");
    payments[id].status = newStatus;
    emit StatusUpdated(id, newStatus);
}

function confirmSettlement(uint256 id) external onlyBank {
    require(payments[id].status == Status.Validated, "Not validated");
    payments[id].status = Status.Settled;
    emit PaymentSettled(id);
}
```

5.3. Block Creation and Consensus Algorithm

After a payment transaction is submitted, the blockchain network checks it. When it follows all the rules, it is placed in a block. The block is then added to the chain using the network's consensus method.

Proof of Authority (PoA)

For a permissioned cross-border payment network, Proof of Authority is a suitable consensus method [3]. Only authorized validator nodes from the participating institutions validate the transactions.

- Only approved validator nodes validate transactions.
- It avoids the heavy mining work of Proof of Work.
- It is suitable for a controlled institutional environment.
- The participating organizations decide which trusted nodes act as validators.

5.4. Oracles

A blockchain oracle gives outside (off-chain) information to a smart contract [6]. A blockchain cannot get such data by itself. In a payment system, an oracle can provide:

- Exchange-rate information
- Bank or payment-system status
- Compliance or verification results
- Transaction reference from an external payment rail
- Current date and time

Note: If the basic project does not need outside data, an oracle is not required. It is an optional extension.

5.5. Deploying and Integrating with a Frontend

After testing on Ganache, the contract is deployed to a test network and then to the permissioned PoA network. The frontend is built with React.js, and Web3.js or Ethers.js is used to connect it to the smart contract. Through simple screens, senders can start payments and view status, and banks can validate payments and confirm settlement. Users do not need to know blockchain technology to use the application.

```
import { ethers } from "ethers";

const provider = new ethers.BrowserProvider(window.ethereum);
const signer = await provider.getSigner();
const contract = new ethers.Contract(contractAddress, abi, signer);

// sender starts a payment
const tx = await contract.initiatePayment(beneficiary, "USD", amount);
await tx.wait();

// read the status of a payment
const status = await contract.getPaymentStatus(paymentId);
```

5.6. Workflow

The workflow of the system is shown in Figure 5.1. The sender logs in, enters the beneficiary details, destination country and amount, and submits the payment. The system checks the details, and the smart contract applies the rules and creates the payment record with the status “Initiated”.

The validators confirm the transaction and it is recorded on the blockchain. The bank then validates the payment and settlement is completed through the normal banking channel. The beneficiary receives the funds, and all authorized parties can view the status. If any detail is not valid, the payment is rejected and the user gets an error message.

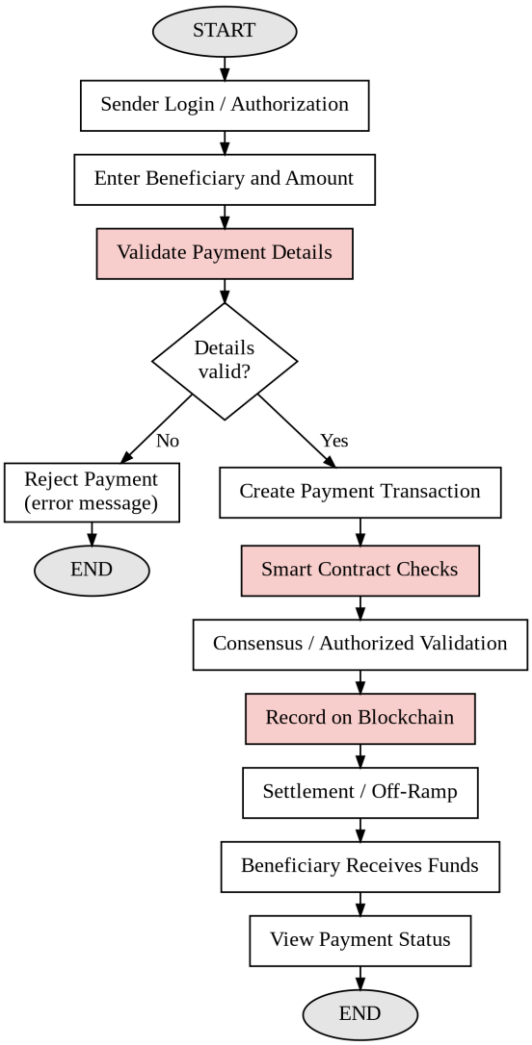


Figure 5.1: Workflow of Blockchain-Based Cross-Border Payments

5.7. Security Aspects

- **Authentication:** Only registered and authorized users can use the system.
- **Roles and permissions:** Sender, bank, administrator and beneficiary have different permissions.
- **Smart contract rules:** The contract checks if the caller is allowed before changing any payment.
- **Tamper resistance:** Recorded transactions are difficult to change later.
- **Traceability:** The full history of a payment can be traced and audited.
- **Private key security:** Blockchain accounts must keep their private keys safe.
- **Input validation:** Beneficiary details, amounts and currency are checked before a transaction is created.
- **Secure external data:** Oracle sources and connections must be trusted and protected.

6. BENEFITS

A blockchain-based cross-border payment system brings several advantages compared with the traditional method.

6.1. Transparency and Auditability

All approved parties see the same record of a payment. This reduces disputes and makes audits easier.

6.2. Better Traceability

The sender, bank and receiver can follow a payment from start to end instead of waiting for manual updates.

6.3. Faster Processing

Smart contracts apply the payment rules automatically, which saves the time taken by manual checking.

6.4. Easier Reconciliation

Because all banks use one shared record, there is less need to match separate ledgers and correct differences.

6.5. Security and Tamper Resistance

Once a transaction is recorded, it is very difficult to change it. Role-based access allows only authorized users to perform payment actions.

6.6. Lower Operational Effort

Less paperwork and fewer manual steps can reduce the cost of handling each payment for banks and providers.

6.7. Controlled Access

A permissioned network lets institutions decide who can join, so sensitive payment data is shared only with trusted members.

6.8. Easy Extension

The system can later be extended with oracles for exchange rates, compliance checks and links to other payment rails.

7. CHALLENGES

Blockchain-based payments also have some limitations that must be considered:

7.1. Scalability

A large payment network can create many transactions, so processing and storage must be planned well.

7.2. Cost and Resources

Blockchain infrastructure and validator nodes need computing resources and maintenance.

7.3. Privacy

Payment information is sensitive, so access to transaction details must be controlled.

7.4. Regulatory and Compliance Requirements

Cross-border payments need identity checks, anti-fraud checks and rules that differ from country to country.

7.5. Data and Input Errors

A blockchain cannot know if the original input was wrong. If wrong details are entered, they will be recorded as they are.

7.6. Key and Account Security

If a private key or an account is compromised, unauthorized actions can take place.

7.7. Smart Contract Logic

A mistake in the contract logic can give wrong results, so the contract must be tested and audited carefully.

7.8. External Data

Exchange rates and bank status come from outside the blockchain, so they need a trusted oracle or integration.

These challenges can be reduced with proper planning, compliance support, security audits and user training.

8. CONCLUSION

This report explained how blockchain can improve cross-border payments. Traditional payments are slow, costly and hard to track because many banks keep separate records. A permissioned blockchain gives all approved participants one shared and tamper-resistant record of each payment. Smart contracts check the rules automatically, Proof of Authority confirms transactions quickly, and the payment status can be followed from start to end.

The implementation described here covers participant registration, payment initiation, status viewing and settlement confirmation using a Solidity smart contract. Security measures such as role-based access, input validation and private key protection keep the system safe. Although challenges like scalability, privacy and regulations exist, they can be managed with careful design.

8.1. Future Outlook for Enhancements

- Adding oracles for live exchange rates and bank status.
- Adding automatic compliance and identity checks.
- Connecting more banks and payment rails to the network.
- Using Layer 2 or other scaling methods to handle more transactions.
- Using Artificial Intelligence to detect unusual or suspicious payments.
- Using decentralized identity (DID) for safer user verification.

With these improvements, blockchain-based cross-border payments can become a secure, clear and widely used solution.

9. SDG's ADDRESSED

The Blockchain-Based Cross-Border Payments system supports the following United Nations Sustainable Development Goals (SDGs) [7]:

SDG 9: Industry, Innovation, and Infrastructure

The system uses blockchain and smart contracts to modernize payment infrastructure. Automated and shared records make financial services more reliable and efficient.

SDG 10: Reduced Inequalities

Faster and clearer international transfers can help people and small businesses, especially those who send or receive money across borders. Lower processing effort can support lower transfer costs.

SDG 16: Peace, Justice and Strong Institutions

A traceable and tamper-resistant payment record supports accountability and helps in detecting fraud. Clear records make financial institutions more transparent and trustworthy.

10. REFERENCES

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11. APPENDIX

Project Repository

GitHub repository: https://github.com/BhavithaAnisetty/crosschainpay.git